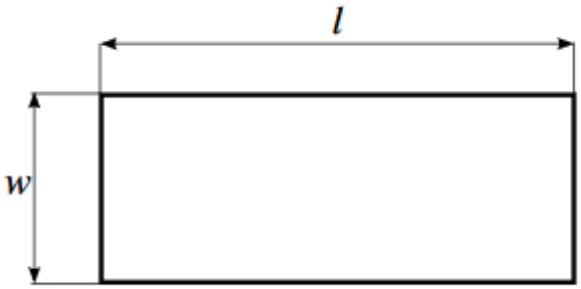
Task 1

Given is an aluminum panel of an aircraft fuselage with the thickness t, framed by stringers and frames on all four sides. They all have a low torsional stiffness.

l = 500 mm
w = 200 mm
t = 2 mm
$\nu = 0.33$



For all following tasks, a constant uniaxial compression load is applied to the shorter sides:

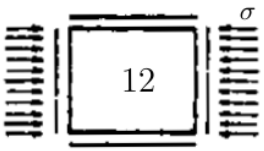
$\sigma_c = 30\ N/mm^2$

Task 1

1 a) Assess the buckling stability of the panel for a uniaxial compression load via buckling diagrams (HSB) by calculating a safety factor.

HSB-45111-01

Boundary condition:





$$\alpha = \frac{l}{w} = 2.5$$

From diagram p.4: $k = 4.1$

Formula from HSB sheet for ν: $k^* = 0.923 \cdot k = 3.784$

Critical buckling stress with formula from HSB sheet: $\sigma_{crit} = k^* \cdot E \cdot \left(\frac{t}{b}\right)^2 = 26.5 \frac{N}{mm^2}$

Assessment: $SF = \frac{\sigma_{crit}}{\sigma_c} = 0.88 < 1$ not stable against buckling

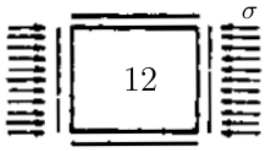
Task 1

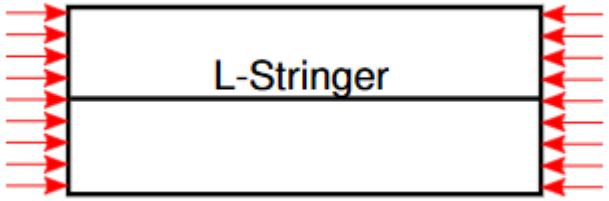
1 b) The panel is now stiffened with an additional stringer in the middle, parallel the two longer sides. The stringer has a L-shaped cross section. Assess the stability against buckling for a uniaxial compression load via buckling diagrams (HSB).

HSB-45111-01

Boundary condition:

L-Stringer low torsion stiffness





$$\alpha = \frac{l}{w/2} = 5$$

From diagram p.4: k converges to value $k_\infty = 4.0$

Formula from HSB sheet for ν: $k^* = 0.923 \cdot k = 3.692$

Critical buckling stress with formula from HSB sheet: $\sigma_{crit} = k^* \cdot E \cdot \left(\frac{t}{b}\right)^2 = 103.4 \frac{N}{mm^2}$

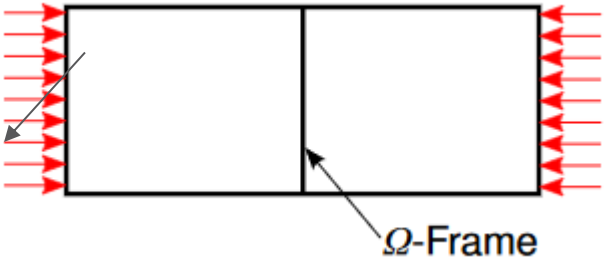
Assessment: $SF = \frac{\sigma_{crit}}{\sigma_c} = 3.45 > 1$ stable against buckling with high safety factor

Task 1

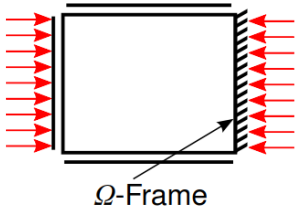
1 c) The initial panel from a) is now stiffened with an additional profile in the middle, parallel to the two shorter sides. The profile has a Ω-shaped cross section. Assess the stability against buckling for a uniaxial compression load via buckling diagrams (HSB). If the BC is ambiguous, compare a conservative and non-conservative approach.

HSB-45111-01

Boundary condition not available
-> case 12 is used because it is conservative



$$\alpha = \frac{l/2}{w} = 1.25$$



From diagram p.4: $k = 4.2$

Formula from HSB sheet for ν: $k^* = 0.923 \cdot k = 3.877$

Critical buckling stress with formula from HSB sheet: $\sigma_{crit} = k^* \cdot E \cdot \left(\frac{t}{b}\right)^2 = 27.1 \frac{N}{mm^2}$

Assessment: $SF = \frac{\sigma_{crit}}{\sigma_c} = 0.90 < 1$ not stable against buckling

In reality, the critical buckling stress will be higher due to the boundary condition.

Task 1

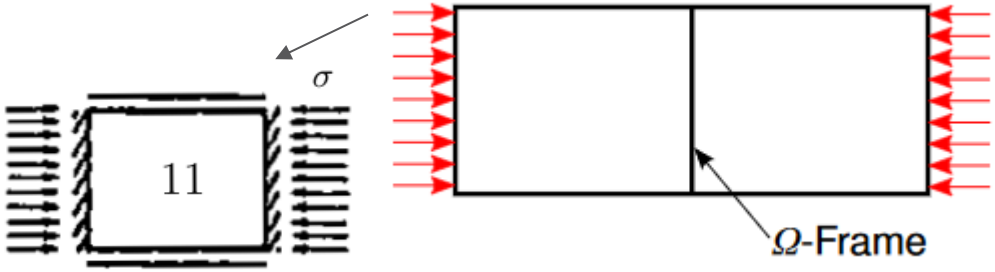
1 c) The initial panel from a) is now stiffened with an additional profile in the middle, parallel to the two shorter sides. The profile has a Ω-shaped cross section. Assess the stability against buckling for a uniaxial compression load via buckling diagrams (HSB). If the BC is ambiguous, compare a conservative and non-conservative approach.

HSB-45111-01

Non-conservative approach:

$\alpha = \frac{l/2}{w} = 1.25$

From diagram p.4 : $k = 5.5$



Formula from HSB sheet for ν: $k^* = 0.923 \cdot k = 5.077$

Critical buckling stress with formula from HSB sheet: $\sigma_{crit} = k^* \cdot E \cdot \left(\frac{t}{b}\right)^2 = 35.5 \ \frac{N}{mm^2}$

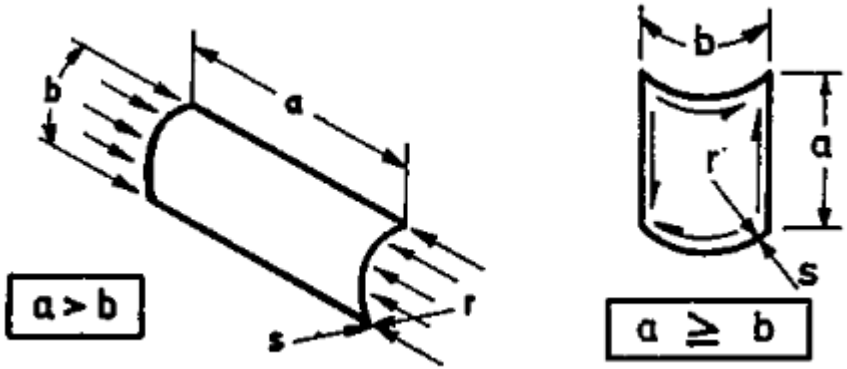
Assessment: $S = \frac{\sigma_{crit}}{\sigma_c} = 1.18 > 1$

According to non-conservative approach barely stable against buckling. True critical buckling stress will be between conservative and non-conservative one and will have very low safety.
-> Overall assessment: Component is not sufficiently buckling resistant.

Task 2

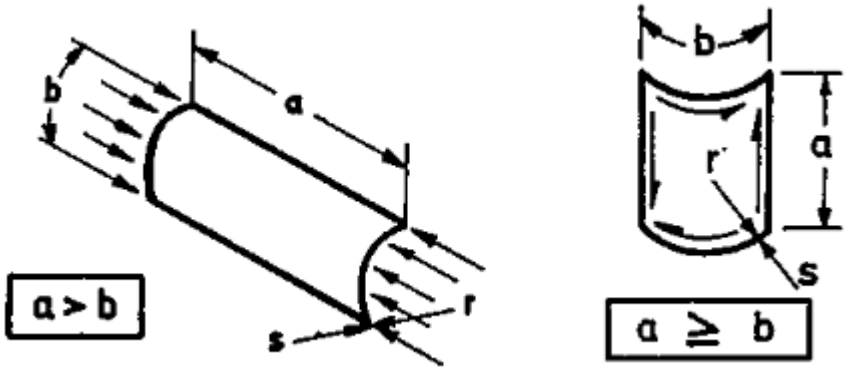
Given is a curved aluminum shell of an aircraft fuselage with the thickness s. The shell is surrounded by profiles with low torsional stiffness.

a = 600 mm
b = 315 mm
s = 1 mm
r = 710 mm
$\nu = 0.4$



Compare the critical stresses of the curved shell with an uncurved shell of the same dimensions for two load cases: compression load and shear load

Task 2



2 a) Load case compression, uncurved shell

HSB-45111-01

$$\alpha = \frac{a}{b} = 1.9$$

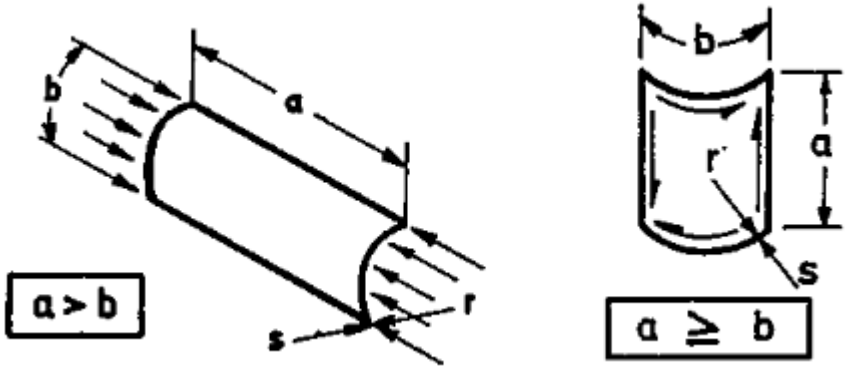
Case 12:

From diagram p.4: $k = 4.0$

Formula from HSB sheet: $\sigma_{crit} = k \cdot \frac{\pi^2 \cdot E}{12 \cdot (1-\nu^2)} \cdot \left(\frac{t}{b}\right)^2 = 2.76 \frac{N}{mm^2}$

Task 2

2 b) Load case compression, curved shell



HSB-45411-01

$a > b$

Diagram p. 3

$\alpha = \frac{a}{b} = 1.9$ Hier: s = t $\frac{r}{t} = 710$

$$\frac{b}{\sqrt{r \cdot t}} \cdot \sqrt[4]{1 - \nu^2} = 11.318$$

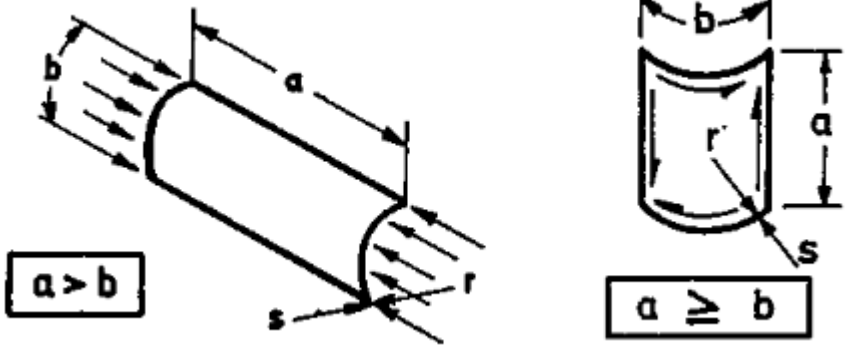
From diagram p.3 for all boundary conditions: $k = 36$

Formula from HSB sheet: $\sigma_{crit} = k \cdot \frac{\pi^2 \cdot E}{12 \cdot (1 - \nu^2)} \cdot \left(\frac{t}{b}\right)^2 = 24.9 \frac{N}{mm^2}$

-> curve has a significant stiffening effect

Task 2

2 c) Load case shear, uncurved shell



HSB-45112-01

$$\frac{b}{a} = 0.525$$

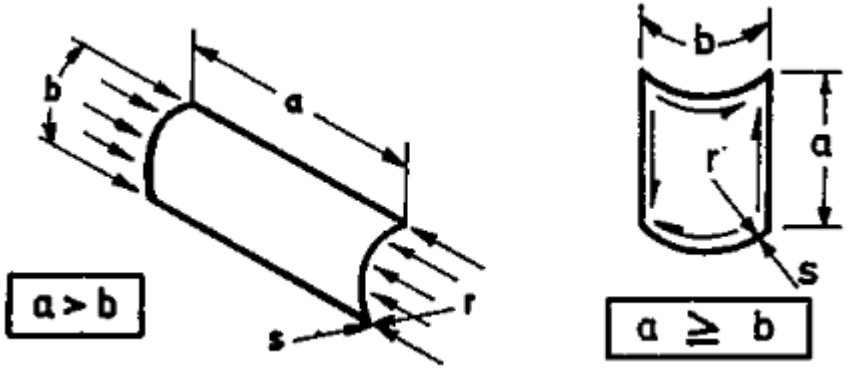
BCs low torsional stiffness

From diagram p.4: $k = 6.5$

Formula from HSB sheet: $\tau_{crit} = k \cdot \frac{\pi^2 \cdot E}{12 \cdot (1-\nu^2)} \cdot \left(\frac{t}{b}\right)^2 = 4.49 \frac{N}{mm^2}$

Task 2

2 c) Load case shear, curved shell



HSB-45412-01

$a > b$ configuration A

$\frac{a}{b} = 1.9$

BCs low torsional stiffness

$$\frac{b}{\sqrt{r \cdot t}} \cdot \sqrt[4]{1 - \nu^2} = 11.318$$

From diagram p.3: $k = 29$

Formula from HSB sheet: $\tau_{crit} = k \cdot \frac{\pi^2 \cdot E}{12 \cdot (1 - \nu^2)} \cdot \left(\frac{t}{b}\right)^2 = 20.03 \frac{N}{mm^2}$

-> curve has a significant stiffening effect